

Group-Balanced Counterfactual Logit Pairing for Robust Toxicity Detection under Demographic Subpopulation Shift

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Abstract

Toxicity detection models can perform well on average while still behaving unevenly across demographic subgroups and language varieties. This paper studies whether identity-term debiasing methods remain reliable under demographic subpopulation shift and whether their fairness gains transfer to a dialect-oriented evaluation setting. I evaluate ERM, CDA, CLP, and a group-balanced CLP variant (GB-CLP) on CivilComments-WILDS using a DistilBERT classifier. GB-CLP combines CLP-style absolute logit pairing with inverse-frequency weighting over label–primary-identity groups, with $\lambda = 1.0$ selected through a validation-only ablation. On the CivilComments test subset, GB-CLP reduces the maximum absolute false-positive-rate gap from 0.1392 for ERM to 0.0440 and gives the highest mean BPSN AUC. However, ERM keeps the best ROC-AUC and macro-F1, CLP gives the best worst-group accuracy, and CDA performs best on the exploratory Davidson proxy dialect FPR gap. These results show a fairness–performance trade-off: counterfactual debiasing can improve selected identity-based fairness measures, but its benefits are metric-dependent and do not automatically transfer to dialectal variation.

Keywords

toxicity detection, unintended bias, counterfactual logit pairing, subpopulation shift, fairness, dialectal variation

1 Introduction

Automatic toxicity detection is often used to support large-scale content moderation, where platforms need to process more comments than human moderators can review manually. Prior work has shown that these systems can perform well on average while still behaving unfairly for comments that mention particular demographic identities [2, 6]. This creates a practical problem because false positives can suppress non-toxic discussion, while false negatives can leave harmful comments visible.

The limitation studied in this paper is not simply that toxicity classifiers can be biased. That issue is already well established. The more specific question is whether debiasing methods remain reliable when the evaluation setting becomes harder. In particular, this work focuses on demographic subpopulation shift and dialectal variation. CivilComments-WILDS is suitable for this setting because it treats identity-defined groups as subpopulations and evaluates robustness under distribution shift [8]. Dialectal variation is also important because prior work has shown that African American English (AAE)-aligned text can be treated differently by hate-speech and toxicity systems [1, 13].

This paper evaluates *Group-Balanced Counterfactual Logit Pairing* (GB-CLP), a simple empirical combination of counterfactual logit

pairing and identity-aware group reweighting. The method uses the absolute logit-pairing loss from counterfactual logit pairing [7] and adds inverse-frequency weights over label–primary-identity groups. The aim is not to claim a new theoretical fairness objective, but to test whether this controlled combination improves subgroup robustness compared with ERM, CDA, and CLP.

The contribution of this paper is empirical and diagnostic. I compare GB-CLP against ERM, CDA, and CLP using the same model, tokenizer, dataset splits, training budget, validation protocol, and thresholding procedure. The evaluation reports both standard predictive metrics and fairness-oriented metrics, including worst-group accuracy, worst-group AUC, Subgroup AUC, BPSN AUC, BNSP AUC, and false-positive-rate gaps. I also include a validation-only ablation for the CLP regularisation strength and a per-identity error analysis across all methods. The results show a fairness–performance trade-off rather than a simple win for one method.

2 Related Work

Toxicity detection has often been studied as large-scale text classification, with early work using crowdsourcing and machine learning to identify personal attacks at scale [15]. Later work showed that toxicity classifiers can learn unintended associations between demographic identity terms and toxicity labels. Dixon et al. [6] studied unintended bias in text classification, while Borkan et al. [2] introduced Subgroup AUC, BPSN AUC, and BNSP AUC to measure identity-related bias beyond average accuracy.

Counterfactual debiasing addresses this problem by testing whether predictions change when identity terms are substituted. Garg et al. [7] proposed counterfactual data augmentation and counterfactual logit pairing, which encourage models to behave consistently under demographic term swaps. Later counterfactual text work shows that the choice of identity substitutions also matters: Wadhwa et al. [14] study identity-pair construction for text fairness, while Lohia [10] considers multi-token counterfactual generation for richer sensitive-attribute changes.

The group-balancing part of this paper is related to reweighting and worst-group robustness. Group DRO directly optimises worst-case group loss under group shifts [11], class-balanced loss reweights examples using the effective number of samples [3], Just Train Twice upweights examples misclassified by an initial ERM model [9], and FORML learns sample weights to balance fairness, robustness, and accuracy [16]. These works show that reweighting is an established idea, so the novelty here is narrow: this study empirically combines CLP with simple identity-aware inverse-frequency weighting in a controlled CivilComments-WILDS toxicity experiment.

Subpopulation shift and dialectal variation provide the evaluation setting. WILDS introduced benchmarks for real-world distribution shifts, including CivilComments-WILDS where demographic identities define subpopulations [8]. Dialectal bias is related but distinct: Blodgett et al. [1] studied AAE-aligned language, while Sap et al. [13] showed that hate-speech systems can reflect racialised dialect bias. Davidson et al. [4] provide the external hate-speech/offensive-language dataset used here for exploratory transfer evaluation.

3 Preliminaries and Problem Definition

Let x_i be a comment and $y_i \in \{0, 1\}$ be its toxicity label. I write $z_\theta(x_i)$ for the model logit and

$$p_\theta(x_i) = \sigma(z_\theta(x_i)) \quad (1)$$

for the predicted probability that x_i is toxic, where $\sigma(\cdot)$ is the sigmoid function. A standard toxicity classifier can be trained using binary cross-entropy:

$$\mathcal{L}_{\text{BCE}}(x_i, y_i) = -y_i \log p_\theta(x_i) - (1 - y_i) \log(1 - p_\theta(x_i)). \quad (2)$$

The main problem is to reduce unfair behaviour under demographic subpopulation shift without losing too much overall predictive quality. In CivilComments-WILDS, identity metadata define demographic subgroups. In the revised experiment, the group used for training weights is

$$g_i = (y_i, s_i), \quad (3)$$

where s_i is the primary identity group assigned to the comment. The possible values are *LGBTQ*, *black*, *christian*, *female*, *male*, *muslim*, *other religions*, *white*, and *no identity*. If no tracked identity is present, s_i is set to *no identity*. If several identities are present, the experiment uses one primary identity according to a fixed metadata order. This avoids the stronger information loss caused by collapsing all identity mentions into a single identity-present indicator, while still keeping the group structure practical for small controlled subsets.

The fairness question is evaluated using both threshold-independent and threshold-dependent measures. Threshold-independent metrics include ROC-AUC, Subgroup AUC, BPSN AUC, BNSP AUC, and worst-group AUC. Threshold-dependent metrics include macro-F1, worst-group accuracy, and false-positive-rate gaps. The classification threshold is selected only on the validation set using macro-F1, then fixed for test evaluation.

4 Proposed Method

The proposed method, GB-CLP, combines two existing ideas: counterfactual logit pairing and group-balanced weighting. For a comment x_i , an identity-swapped counterfactual x'_i is generated by replacing explicit identity terms using transparent swap pairs, such as *muslim/christian*, *black/white*, and *woman/man*. Following the CLP objective in Garg et al. [7], the pairing term uses the absolute difference between the original and counterfactual logits:

$$\mathcal{L}_{\text{pair}}(x_i, x'_i) = |z_\theta(x_i) - z_\theta(x'_i)|. \quad (4)$$

This loss encourages the model to keep similar logit scores when only the explicit identity term has been changed. The absolute difference is used instead of squared error so that the implementation

matches the CLP-style robustness objective rather than introducing a separate MSE-based variant.

The second idea is group-balanced weighting. Let $\mathcal{D}_{\text{train}}$ be the training set and let n_g be the number of training examples in group g . Each example receives an inverse-frequency weight w_{g_i} based on its label–primary-identity group. The raw weight is proportional to $1/n_{g_i}$ and is normalised to have mean one across the training set. This keeps the overall loss scale stable while increasing the contribution of underrepresented label–identity groups.

For a mini-batch $\mathcal{B}$, let $|\mathcal{B}|$ be the mini-batch size and let $m_i \in \{0, 1\}$ indicate whether the counterfactual generation actually changed the text. Define $M_{\mathcal{B}} = \sum_{i \in \mathcal{B}} m_i$. The batch-level pairing term is

$$\mathcal{L}_{\text{pair}}^{\mathcal{B}} = \begin{cases} \frac{1}{M_{\mathcal{B}}} \sum_{i \in \mathcal{B}} m_i \mathcal{L}_{\text{pair}}(x_i, x'_i), & M_{\mathcal{B}} > 0, \\ 0, & M_{\mathcal{B}} = 0. \end{cases} \quad (5)$$

The second case is only an implementation safeguard: if a mini-batch has no changed counterfactual pairs, the pairing term contributes zero rather than causing division by zero.

The final GB-CLP loss is

$$\mathcal{L}_{\text{GB-CLP}} = \frac{1}{|\mathcal{B}|} \sum_{i \in \mathcal{B}} w_{g_i} [\mathcal{L}_{\text{BCE}}(x_i, y_i) + \mathcal{L}_{\text{BCE}}(x'_i, y_i)] + \lambda \mathcal{L}_{\text{pair}}^{\mathcal{B}}, \quad (6)$$

where λ controls the strength of the counterfactual consistency penalty. A validation-only ablation is used to select λ before test evaluation. This design directly tests whether CLP-style identity invariance, when combined with simple identity-aware group reweighting, improves robustness under subpopulation shift.

The method is intentionally simple. This makes it easier to compare against ERM, CDA, and CLP under the same experimental protocol, but it also means the contribution is an empirical adaptation rather than a new theoretical fairness algorithm.

5 Experimental Setup

5.1 Datasets

The main benchmark is CivilComments-WILDS [8]. The experiment uses the official train, validation, and test splits rather than a random split because the goal is to preserve the benchmark’s subpopulation-shift structure. Controlled subsets were sampled from the official splits: 50,000 training examples, 10,000 validation examples, and 20,000 test examples. Sampling was stratified by toxicity label and primary identity group, rather than only by whether any identity was present.

For dialect-transfer evaluation, the Davidson hate-speech and offensive-language dataset was used as an external evaluation dataset [4]. Hate-speech and offensive-language labels were mapped to the toxic class, while the neither class was mapped to the non-toxic class. The notebook used 10,000 examples. Because no validated dialect probability file was used in the current experiment, the dialect analysis is reported only as an exploratory proxy evaluation. The two groups are labelled *AAE_proxy* and *Other_proxy*; they should not be interpreted as confirmed dialect labels.

5.2 Baselines and Model

All methods use *distilbert-base-uncased*, a distilled version of BERT [5, 12]. The four compared methods are:

- **ERM**: standard empirical risk minimisation with binary cross-entropy.
- **CDA**: counterfactual data augmentation using identity-swapped examples.
- **CLP**: counterfactual logit pairing with original and identity-swapped examples using the absolute logit-difference penalty.
- **GB-CLP**: CLP with inverse-frequency weighting over label-primary-identity groups.

All methods used the same model backbone, tokenizer, maximum sequence length, optimiser, validation protocol, and thresholding procedure. Unless otherwise stated, experiments used a maximum sequence length of 192, batch size 64, AdamW with learning rate 2×10^{-5} , early-stopping patience 3, and random seed 42 for reproducibility. Models were trained for up to 10 epochs with early stopping. The checkpoint with the highest validation ROC-AUC was selected for final test evaluation. The classification threshold was then chosen on the validation set by maximising macro-F1 and was not tuned on the test set. For CLP and GB-CLP, the pairing strength λ was selected using a validation-only ablation for GB-CLP and then fixed at $\lambda = 1.0$ for the final comparison.

5.3 Evaluation Metrics

Overall predictive performance is measured using ROC-AUC, macro-F1, accuracy, precision, and recall. Fairness and robustness are measured using worst-group accuracy, worst-group AUC, mean Subgroup AUC, mean BPSN AUC, mean BNSP AUC, and the maximum absolute false-positive-rate gap across identity groups. Subgroup AUC, BPSN AUC, and BNSP AUC follow the unintended-bias evaluation framework introduced by Borkan et al. [2].

6 Results

6.1 CivilComments-WILDS Results

Table 1 reports the main CivilComments test results after using absolute CLP loss and label-primary-identity group balancing. The proposed method does not achieve the best value on every metric. ERM has the highest ROC-AUC at 0.9342 and the highest macro-F1 at 0.7935. CLP has the best worst-group accuracy at 0.4806, although it also has the largest maximum absolute FPR gap at 0.3016. The proposed method has the highest accuracy at 0.9162, the highest mean BPSN AUC at 0.8945, and the lowest maximum absolute FPR gap at 0.0440. This suggests that GB-CLP mainly improves false-positive-rate disparity and BPSN-style separation, rather than producing a universal improvement across all robustness metrics.

Figure 1 shows that all methods remain far above the random baseline. However, the ranking performance is not improved by the debiasing objectives: ERM remains strongest on ROC-AUC, and the proposed method is below ERM and CDA on this metric.

The FPR-gap result in Figure 2 is the clearest positive result for the proposed method. Compared with ERM, GB-CLP reduces the maximum absolute FPR gap from 0.1392 to 0.0440. Compared with CLP, the reduction is larger, from 0.3016 to 0.0440. The BPSN result in Figure 3 is also favourable for GB-CLP, with mean BPSN AUC increasing to 0.8945. However, the lower BNSP AUC and worst-group accuracy show that this improvement is not uniform across all unintended-bias metrics.

The worst-group accuracy and worst-group AUC columns in Table 1 show why the result should be interpreted carefully. CLP improves worst-group accuracy but produces the largest FPR gap. ERM has the best worst-group AUC, while the proposed method gives the lowest FPR gap. Therefore, the experiment does not support a broad claim that one debiasing method is best in all senses. Instead, the evidence suggests that different fairness metrics capture different behaviours.

6.2 Training Behaviour and λ Selection

The selected validation checkpoints were early in training: epoch 2 for ERM, epoch 1 for CDA, epoch 2 for CLP, and epoch 3 for the proposed method. Training continued for later epochs until early stopping, but validation ROC-AUC did not consistently improve after these points. This supports the use of validation-based checkpoint selection rather than simply using the final epoch.

Table 2 reports a validation-only ablation for the CLP regularisation strength in GB-CLP. Although $\lambda = 2.0$ gives the lowest validation FPR gap, it also reduces ROC-AUC and macro-F1. I therefore select $\lambda = 1.0$ because it gives the best validation macro-F1 while still substantially reducing the FPR gap. This matches the aim of studying fairness-performance trade-offs rather than optimising only one fairness metric.

6.3 Dialect-Transfer Evaluation

Table 3 reports the external Davidson transfer results. The dialect grouping is an exploratory proxy, so the results should be interpreted as a stress test rather than a validated measurement of AAE or White-aligned language. In this setting, ERM has the highest ROC-AUC and macro-F1. CDA gives the lowest proxy dialect FPR gap at 0.0045. The proposed method has a proxy FPR gap of 0.0442, which is much lower than ERM but higher than CDA and CLP. This indicates that the proposed CivilComments FPR-gap improvement does not fully transfer to the dialect-proxy setting.

Figure 4 supports the main generalisation finding. The proposed method improves one CivilComments fairness measure, but it does not automatically produce the best proxy dialect-transfer gap. This is consistent with the concern that identity-term debiasing and dialectal robustness are related but not identical problems.

6.4 Per-Identity Error Analysis

Table 4 summarises the highest false-positive and false-negative identity groups for each method. This table is included because collapsing all demographic attributes into a single identity-present value can hide which identities are most affected. ERM has the highest FPR on the *black* group, while CDA, CLP, and GB-CLP have the highest FPR on the *white* group. The proposed method lowers the worst observed FPR to 0.0851, compared with 0.1869 for ERM, 0.1835 for CDA, and 0.3465 for CLP. However, it also has the highest FN rate for the *LGBTQ* group at 0.6429. This confirms that reducing false positives does not automatically reduce false negatives.

Manual inspection of high-confidence false positives and false negatives also suggests that some errors are not caused only by explicit identity terms. Several high-confidence false positives contain insults or aggressive language without tracked identity mentions,

Table 1: CivilComments-WILDS test results. Higher is better for ROC-AUC, macro-F1, accuracy, worst-group accuracy, worst-group AUC, Subgroup AUC, BPSN AUC, and BNSP AUC. Lower is better for maximum absolute FPR gap.

Method	ROC-AUC	Macro-F1	Acc.	Worst Acc.	Worst AUC	Subgroup AUC	BPSN AUC	BNSP AUC	Max $ \Delta\text{FPR} $
ERM	0.9342	0.7935	0.9148	0.4672	0.8280	0.8723	0.8682	0.9379	0.1392
CDA	0.9297	0.7867	0.9141	0.4444	0.8223	0.8678	0.8653	0.9328	0.1419
CLP	0.9184	0.7881	0.9098	0.4806	0.7949	0.8527	0.8195	0.9354	0.3016
GB-CLP	0.9245	0.7861	0.9162	0.3333	0.8247	0.8691	0.8945	0.9042	0.0440

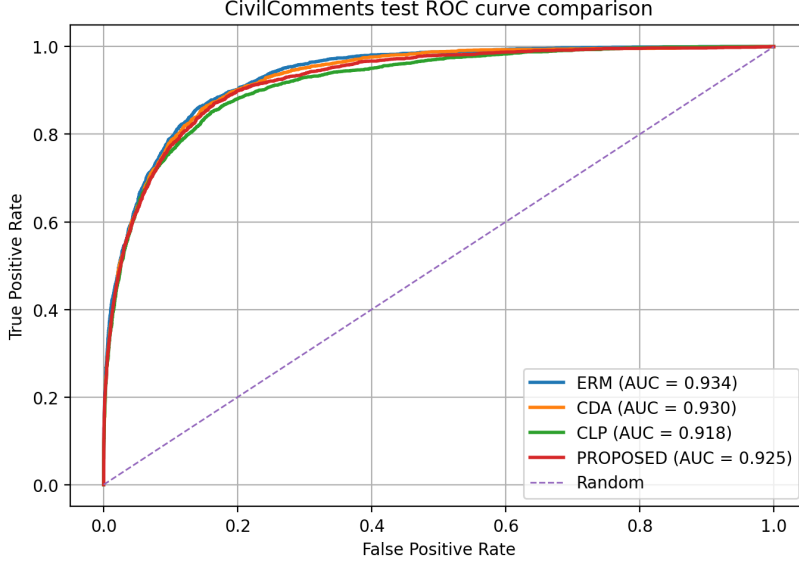


Figure 1: ROC curve comparison on the CivilComments-WILDS test subset.

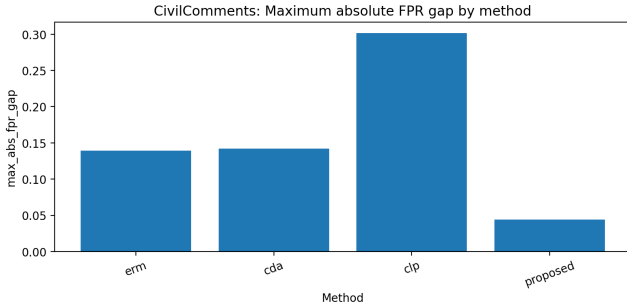


Figure 2: Maximum absolute false-positive-rate gap on CivilComments-WILDS.

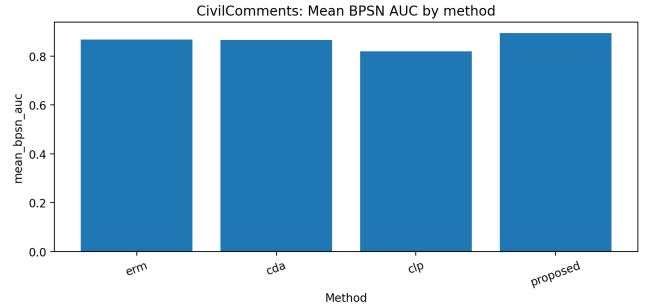


Figure 3: Mean BPSN AUC on CivilComments-WILDS.

Table 2: Validation-only ablation for the CLP regularisation strength in GB-CLP.

λ	ROC-AUC	Macro-F1	Worst Acc.	Worst AUC	Max $ \Delta\text{FPR} $
0.05	0.9169	0.7738	0.2778	0.8007	0.1058
0.10	0.9176	0.7735	0.2778	0.8037	0.0928
0.50	0.9216	0.7808	0.2407	0.8102	0.0981
1.00	0.9197	0.7821	0.2593	0.7964	0.0784
2.00	0.9140	0.7768	0.2963	0.7954	0.0551

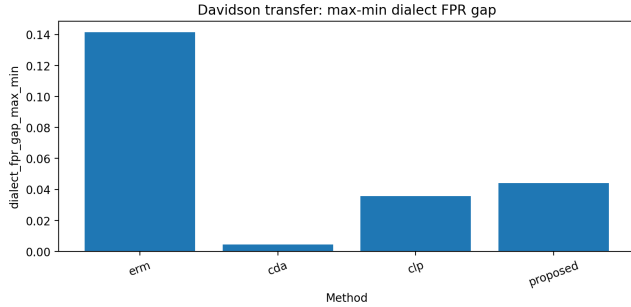
while several false negatives contain sarcastic, indirect, or context-dependent toxicity. This matters because counterfactual identity swapping mainly targets explicit identity terms and is less able to address pragmatic or discourse-level toxicity.

7 Discussion

The main finding is that GB-CLP improves some fairness measures but not all of them. On CivilComments-WILDS, it produces the

Table 3: Davidson transfer evaluation using exploratory proxy dialect groups. Higher is better for ROC-AUC and macro-F1. Lower is better for FPR gap.

Method	ROC-AUC	Macro-F1	Proxy FPR gap
ERM	0.8748	0.7129	0.1415
CDA	0.8546	0.6938	0.0045
CLP	0.8375	0.6904	0.0358
GB-CLP	0.8614	0.6972	0.0442

**Figure 4: Exploratory Davidson transfer false-positive-rate gap.****Table 4: Highest false-positive and false-negative identity groups by method.**

Method	Highest FP group	FP rate	Highest FN group	FN rate
ERM	black	0.1869	christian	0.5135
CDA	white	0.1835	christian	0.4973
CLP	white	0.3465	LGBTQ	0.4835
GB-CLP	white	0.0851	LGBTQ	0.6429

lowest maximum absolute FPR gap and the highest mean BPSN AUC. This provides evidence that identity-aware group-balanced counterfactual training can reduce one form of identity-related false-positive disparity in this controlled setting. The likely reason for this reduction is that the pairing loss discourages large logit changes under identity substitutions, while the identity-aware weights increase the influence of smaller label-identity groups that would otherwise contribute less to the training objective. However, the method has lower worst-group accuracy than ERM, CDA, and CLP, and lower ROC-AUC and macro-F1 than ERM. Therefore, GB-CLP should be interpreted as improving a specific fairness trade-off rather than as a uniformly stronger model.

The results also show that counterfactual debiasing can behave differently depending on the metric. CLP improves worst-group accuracy but increases the maximum absolute FPR gap. CDA performs slightly worse than ERM on the main CivilComments predictive metrics but gives the lowest proxy dialect-transfer FPR gap. GB-CLP strongly reduces CivilComments FPR disparity but increases false negatives for some identity groups. These differences suggest that fairness evaluation should not rely on a single number.

The dialect-transfer result is especially important for interpreting generalisation. GB-CLP was designed around explicit identity-term counterfactuals and label-primary-identity group balancing, while the Davidson proxy evaluation reflects a different source of variation. Since GB-CLP does not clearly achieve the best proxy dialect FPR gap, the experiment supports a limited conclusion: identity-term debiasing may help with some identity-based fairness measures, but it should not be assumed to solve dialectal robustness without direct evidence from a validated dialect evaluation.

From a practical moderation perspective, these results suggest that model selection depends on the type of harm a system is trying to reduce. If the priority is reducing false-positive-rate disparity across CivilComments identity groups, GB-CLP is the strongest option in this experiment because it gives the lowest maximum absolute FPR gap. However, if the priority is worst-group accuracy or overall ranking performance, CLP or ERM may be preferable. If the priority is reducing proxy dialect FPR gap, CDA is strongest in this experiment. This is why the contribution should be read as a diagnostic fairness study rather than a claim that one method is universally best.

8 Limitations

This study has several limitations. First, the experiment uses a controlled subset of CivilComments-WILDS rather than the full dataset, and the reported results come from one random seed. Therefore, the results should be interpreted as controlled empirical evidence rather than a complete benchmark claim, and no statistical significance or confidence intervals are claimed. Second, the dialect-transfer analysis uses an exploratory lexical proxy rather than a validated dialect model, so the dialect results should not be treated as a definitive evaluation of AAE robustness.

Third, the identity-swapping procedure is transparent but simple. Some substitutions may create unnatural text or remove important social context, and explicit swaps do not capture all forms of demographic or dialectal variation. Fourth, all methods use the same DistilBERT backbone. This improves experimental control, but it does not show whether the same trends would hold for larger models or more recent language encoders. Finally, the revised weighting strategy uses primary identity groups rather than a single identity-present indicator, but this is still a simplification. Some comments mention multiple identities, and assigning one primary identity does not fully capture intersectional or multi-identity harms.

9 Conclusion and Future Work

This paper evaluated whether debiasing methods for toxicity detection remain effective under demographic subpopulation shift and whether their gains transfer to a dialect-oriented evaluation setting. The proposed GB-CLP method is an empirical combination of CLP-style absolute logit pairing and inverse-frequency weighting over label-primary-identity groups. On CivilComments-WILDS, GB-CLP reduced the maximum absolute FPR gap and improved mean BPSN AUC while keeping macro-F1 close to the baselines. However, ERM retained the best ROC-AUC and macro-F1, CLP achieved the best worst-group accuracy, and CDA performed best on the exploratory Davidson proxy FPR gap. These results indicate

a fairness–performance trade-off rather than a single dominant method.

The main conclusion is therefore cautious. Group-balanced counterfactual training can improve selected fairness measures in an identity-based subpopulation-shift setting, but the improvement is metric-dependent and does not automatically generalise to the exploratory dialect-transfer setting. Future work should repeat the experiment with multiple random seeds, use the full CivilComments-WILDS dataset, include a validated dialect probability model, and test whether the same trade-offs hold for larger transformer backbones. It would also be useful to design counterfactuals that preserve more linguistic and social context rather than relying only on simple identity-term swaps, and to compare simple inverse-frequency weighting against stronger group-robust methods such as group DRO or learned reweighting.

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